

Of course 

Here's a **complete, clean, structured list** of **all the APIs, programming languages, frameworks, libraries, and models** used in your **MahaKrishi Project** 

1. Programming Languages

Language	Usage
 Python	Backend logic, machine learning models, API creation (Flask).
 JavaScript (React.js)	Frontend development, routing, UI interaction.
 HTML & CSS (via Tailwind CSS)	Styling and structure of the frontend.

2. Backend Stack (APIs & Frameworks)

Technology	Purpose
Flask (Python)	Main backend framework to build RESTful APIs.
Flask-CORS	Enables communication between frontend (React) and backend (Flask).
SQLite3	Lightweight database to store region, market, commodity data.
Ultralytics YOLOv8	Pest detection model integration.
Scikit-learn	Crop recommendation model (Random Forest Classifier).
pandas	Data handling and preprocessing.
requests	Calling external APIs (Weather & News).
logging	Server-side error and info logging.

3. Frontend Stack (Languages & Libraries)

Technology	Purpose
React.js	Core frontend library for building UI.
React Router DOM	Client-side routing between different pages.
Axios / Fetch API	Making API requests to the Flask backend.
Tailwind CSS	Modern utility-first CSS framework for styling.

Technology	Purpose
Recharts	Visualization library to display graphs (crop recommendation chart).

✦ 4. Machine Learning Models

Model	Library	Purpose
RandomForestClassifier	scikit-learn	Crop Recommendation based on N, P, K, temperature, humidity, pH, rainfall.
YOLOv8 (Ultralytics)	PyTorch backend	Pest detection model (pest_best.pt) trained on 12 pest classes.

✅ **Pest classes:** Ants, Bees, Beetles, Caterpillars, Earthworms, Earwigs, Grasshoppers, Moths, Slugs, Snails, Wasps, Weevils.

☁ 5. External APIs Used

API Name	Purpose	Endpoint Used
 OpenWeather API	Fetching live temperature, humidity, weather condition, wind speed.	https://api.openweathermap.org/data/2.5/weather
 News API	Fetching real-time agriculture-related news from trusted sources.	https://newsapi.org/v2/everything

🧱 6. Internal API Endpoints (Flask)

Endpoint	Method	Purpose
/api/weather	GET	Get live weather info for given city.
/api/farmer-news	GET	Get agriculture-related news.
/api/divisions	GET	Get list of divisions (region).
/api/districts	GET	Get list of districts under selected division.
/api/markets	GET	Get markets under selected district.
/api/commodities	GET	Get list of commodities.
/api/prices	GET	Fetch price trends for selected commodity and market.

Endpoint	Method	Purpose
/api/recommend-crops	POST	Get recommended crop using ML model.
/api/pest-detect	POST	Detect pest using YOLOv8 model from uploaded image.
/	GET	Default test route for server running.

7. Database

Database Type	Usage
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SQLite	Relational Stores regions, districts, markets, commodities, and prices.
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Tables in database:

- divisions
- districts
- markets
- commodities
- market_prices

8. Tools & Utilities

Tool / Library	Purpose
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Torch (PyTorch)	Required backend for YOLOv8 model loading.
Ultralytics	High-level API for YOLO object detection models.
pandas	CSV reading and data preprocessing.
sklearn	Machine learning model for crop recommendation.
logging	Backend logging for debugging and monitoring.
os / subprocess	File handling and system-level functions.

9. Folder / File Structure (Simplified)

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└─ Project Root
    └─ Backend
        └─ app.py ← Main Flask application
        └─ models
```

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| | └─ pest_best.pt
| └─ ── data
| | └─ Crop_recommendation.csv
| └─ ── db
| | └─ maha_krishi.db
| └─ ── tmp
| | └─ (uploaded pest images)
| └─ chat.py ← LLM chatbot routes
|
└─ ── Frontend
| └─ src
| | └─ App.js
| | └─ ── pages
| | | └─ Dashboard.js
| | | └─ CropRecommendation.js
| | | └─ PestDetection.js
| | | └─ MarketPrices.js
| | | └─ Chat.js
| | └─ index.js
| | └─ styles.css (Tailwind)
| └─ package.json
|
└─ requirements.txt
└─ README.md

```

10. Tech Stack Summary Table

Category	Technologies Used
Languages	Python, JavaScript (React), HTML, CSS
Frontend	React.js, Tailwind CSS, Recharts, Axios

Category	Technologies Used
Backend	Flask, Flask-CORS, SQLite, Requests, Logging
Machine Learning	YOLOv8 (Ultralytics), PyTorch, scikit-learn, pandas
External APIs	OpenWeather API, News API
Database	SQLite
Other Tools	os, subprocess, logging

✅ In short:

- 🖥️ **Frontend:** React + Tailwind + Recharts
 - 🧠 **Backend:** Flask + ML + YOLO + SQLite
 - 🌱 **Models:** Random Forest for crop recommendation, YOLOv8 for pest detection
 - ☁️ **External APIs:** OpenWeather + News API
 - 🧱 **Internal APIs:** Weather, News, Market, Crop Recommendation, Pest Detection
 - 💾 **Database:** SQLite
 - ⚙️ **Support Libraries:** pandas, scikit-learn, torch, ultralytics, requests
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This list is **everything that your project uses**, covering:

- 💻 Languages
- ⚙️ Frameworks
- 🌐 APIs
- 🧠 ML Models
- 💾 Databases
- 🧰 Libraries & Tools

Perfect to **present as your Tech Stack explanation** during your viva or presentation.

Would you like me to also give you a **1-minute short oral explanation** of this tech stack (like how to say it smoothly to your professor)? 🎤 ✨